03 21826 Autumn 2025
LH The Life and Death of Stars

Lecture 3

(JCD 2)

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Previous lecture

- Hertzsprung-Russell diagram shows evolution of a star
- Effective temperature of a star defined through its luminosity and radius:

$$L = 4\pi\sigma R^2 T_{\text{eff}}^4$$

- Throughout their life stars move through the HR diagram
- Stellar evolution is mass dependent
- Stellar modeling relies on many simplifications

Slido

Join SLIDO - #1098676

Outline

Position and distance

Brightness

Colour-magnitude diagram

Spectral properties

Masses, radii, and ages

Summary

Stellar properties

- Observable properties:
  - Electromagnetic radiation
  - Effect of gravitational field
- Limitation of stellar observations:
  - Most stars are point sources (what is the obvious exception?)
  - Can't "sample" the interior
  - Interstellar dust contaminates stellar light

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Position

- Measure in degrees (or hours), arcminutes, arcseconds
- Right ascension and declination are the celestial sphere equivalent of longitude and latitude
- Many other systems (ecliptic, Galactic, ...)

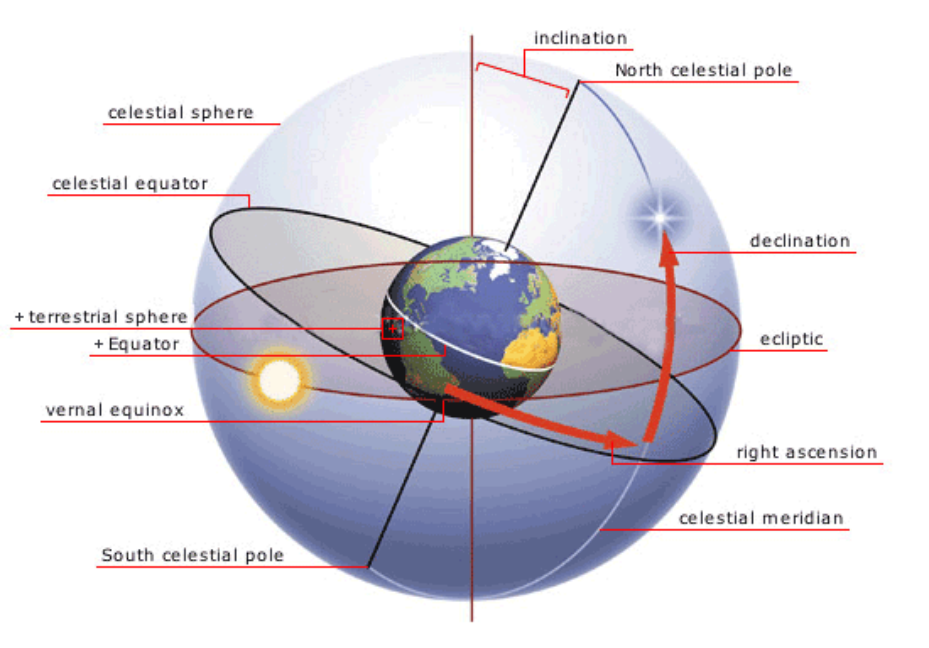

Image credit: Astronomy&Space

Aside: Astrometry as an exoplanet detection technique

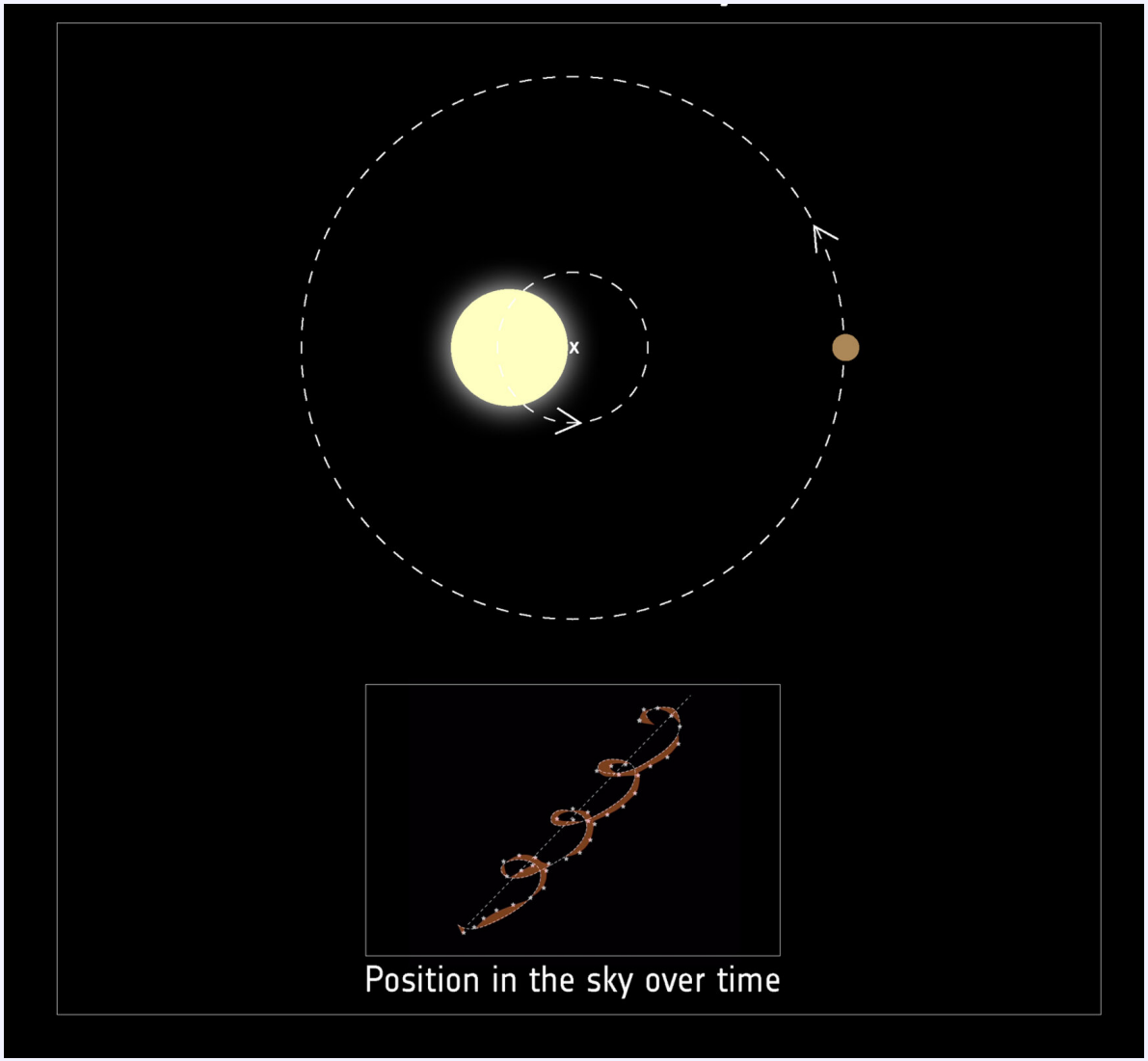

Image credit: ESA

Gaia predicted to find 10k - 50k long-period Jupiters by the end of its mission

Distance

- Measure change in direction to star when Earth goes around the Sun:

$$\tan p = \frac{1\text{AU}}{d}$$

- p is parallax
- Apply small angle formula:

$$d = \frac{1\text{AU}}{p/rad}$$

- Parsec is a unit of distance defined as the distance that creates 1 arcsecond of parallax

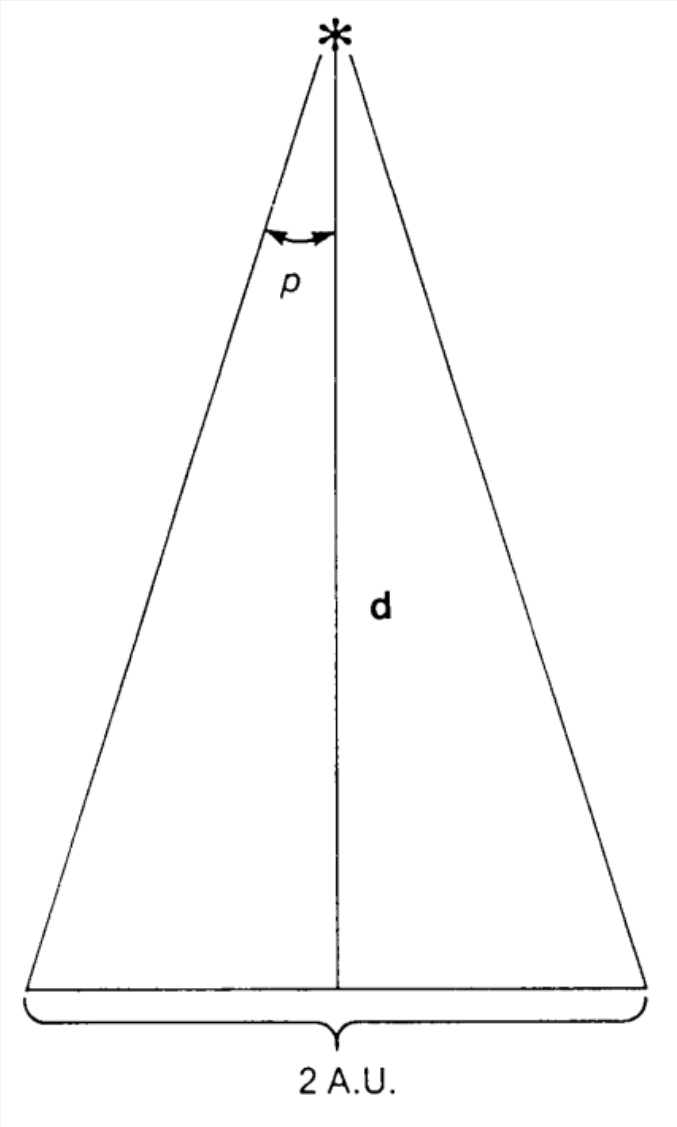

Image credit: JCD

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Luminosity and magnitude

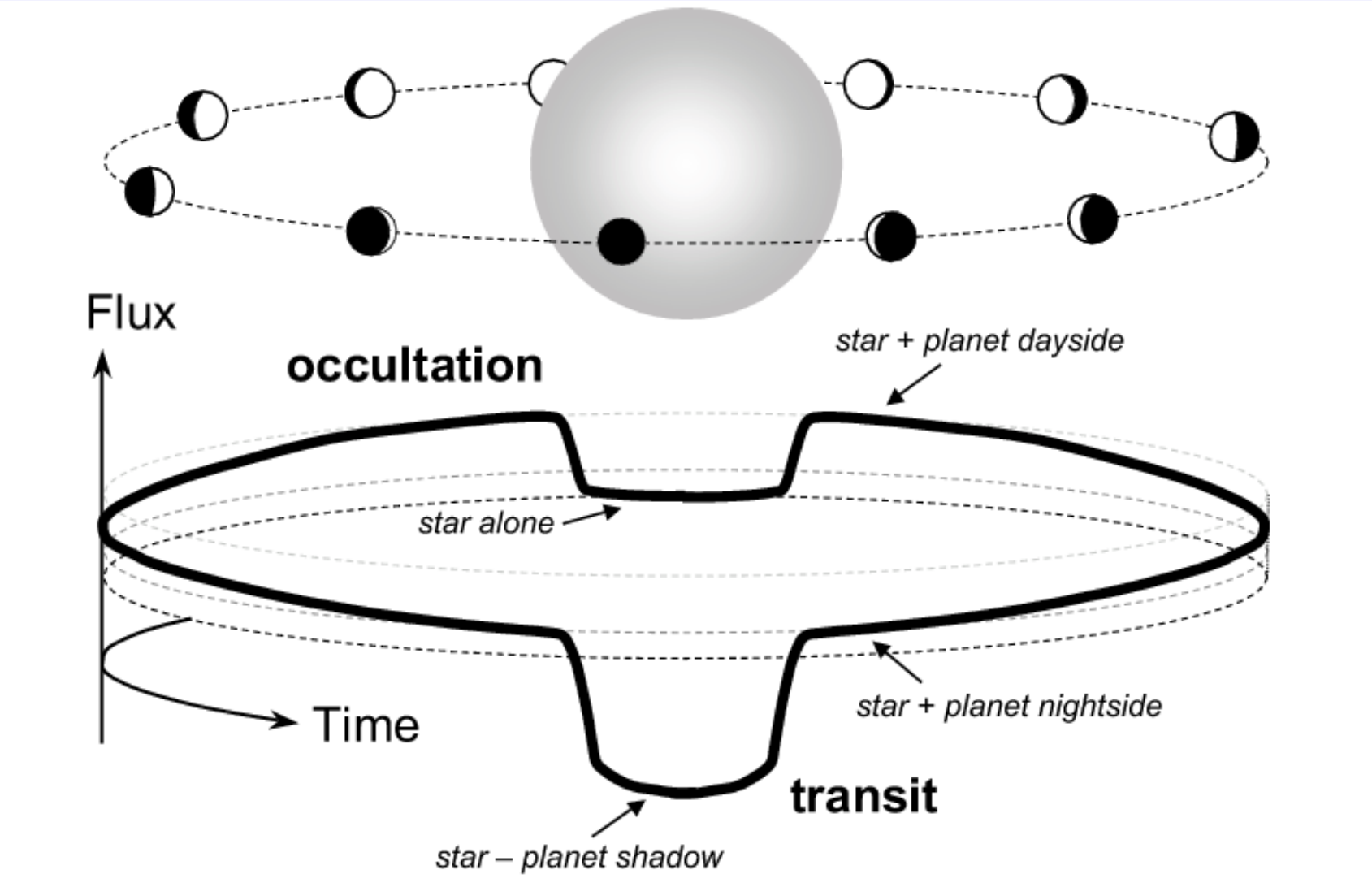
- We measure apparent stellar luminosity, l. This is the stellar energy passing through a unit area in a unit time. $[l] = \mathrm{erg\,cm^{-2}\,s^{-1}}$
- Transform into apparent magnitude: $m = -2.5 \log l + K_1$. Fainter stars have higher magnitudes!
- For stellar evolution, we need the absolute luminosity (L) and magnitude (M), found by including stellar distance:

$$L = 4\pi d^2 l$$

$$M = -2.5 \log L + K_2$$

- Combining that gives $m = M + 5 \log d - 5$ with d measured in parsecs

Aside: brightness/photometry as an exoplanet detection technique



Winn 2011

Colours

- Stellar energy and luminosity is distributed across wavelengths.
- We observe stars in different passband filters
- Stellar colour is defined as the difference in apparent magnitude of a star through different filters, eg U-B, B-V, ...

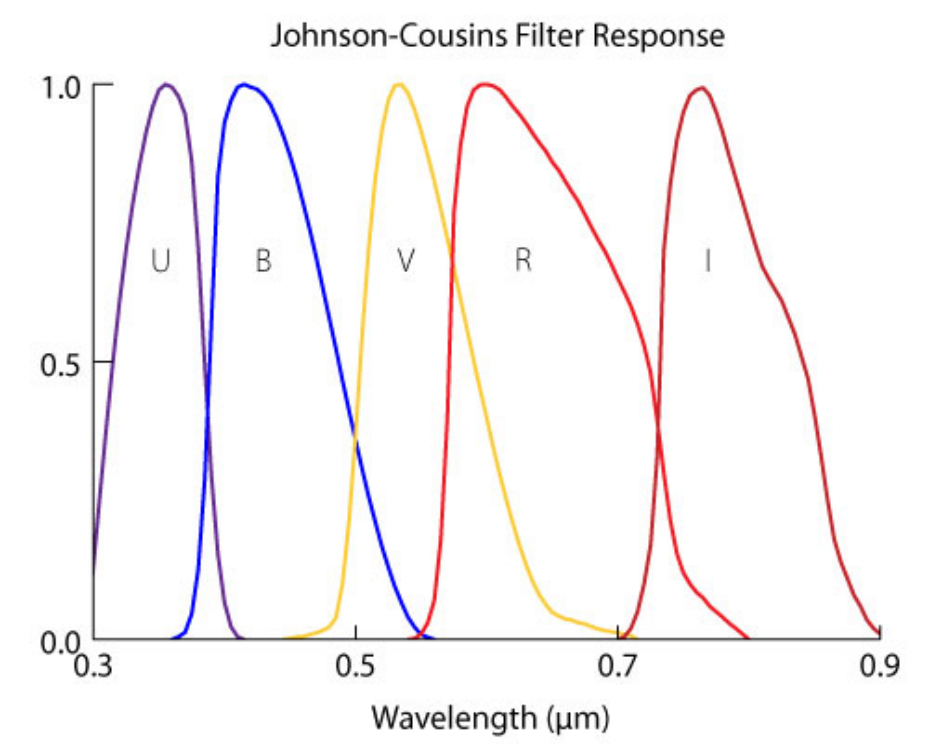


Image credit: Astrophysics Spectator

Effective temperature

- Effective temperature is related to brightness:

$$L = 4\pi\sigma R^2 T_{\text{eff}}^4$$

- Hence colour is related to effective temperature - not 1-1 but close.
- Remember: Effective temperature $\neq$ Internal temperature

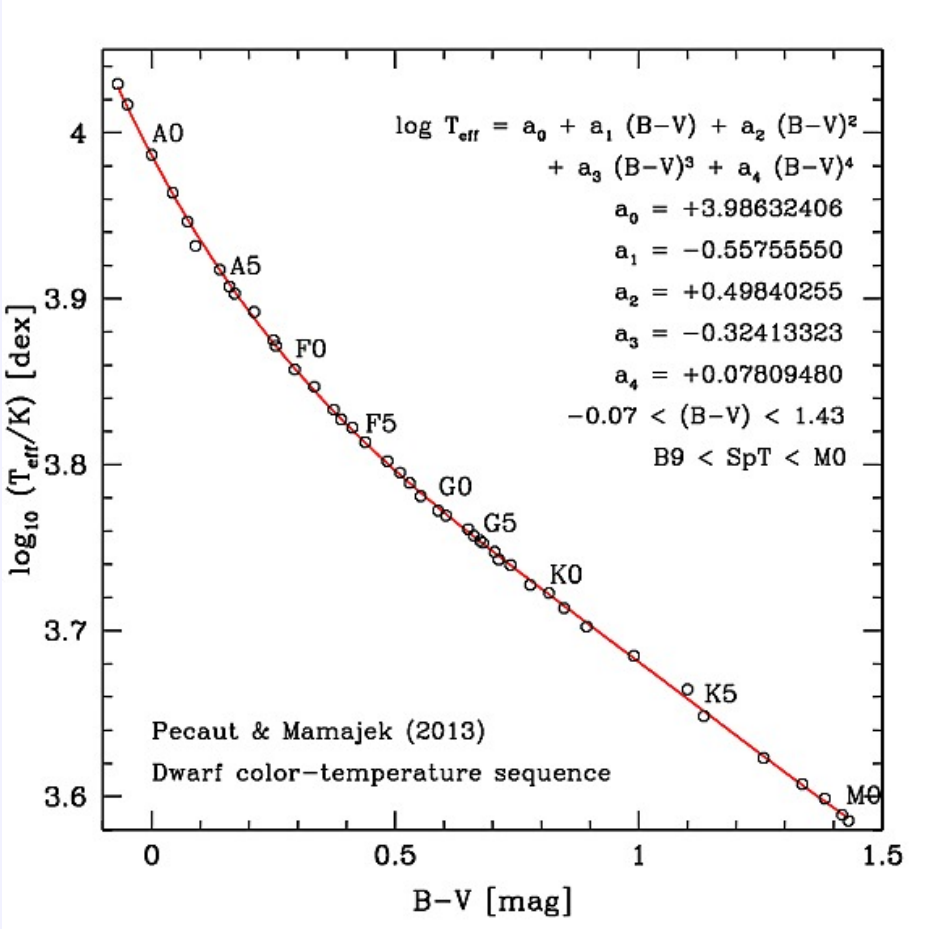


Image credit: E. Mamajek

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Colour-magnitude diagram

- Colour is related to effective temperature
- Absolute magnitude is related to luminosity
- A colour-magnitude diagram is thus an observational HR diagram!
- Need at least 2 apparent magnitudes and a distance in order to make it

Hipparcos

- 1987 - 1993
- $\sim$ 120000 stars with positions within 0.001”
- Less than 20000 stars with distances better than 10%

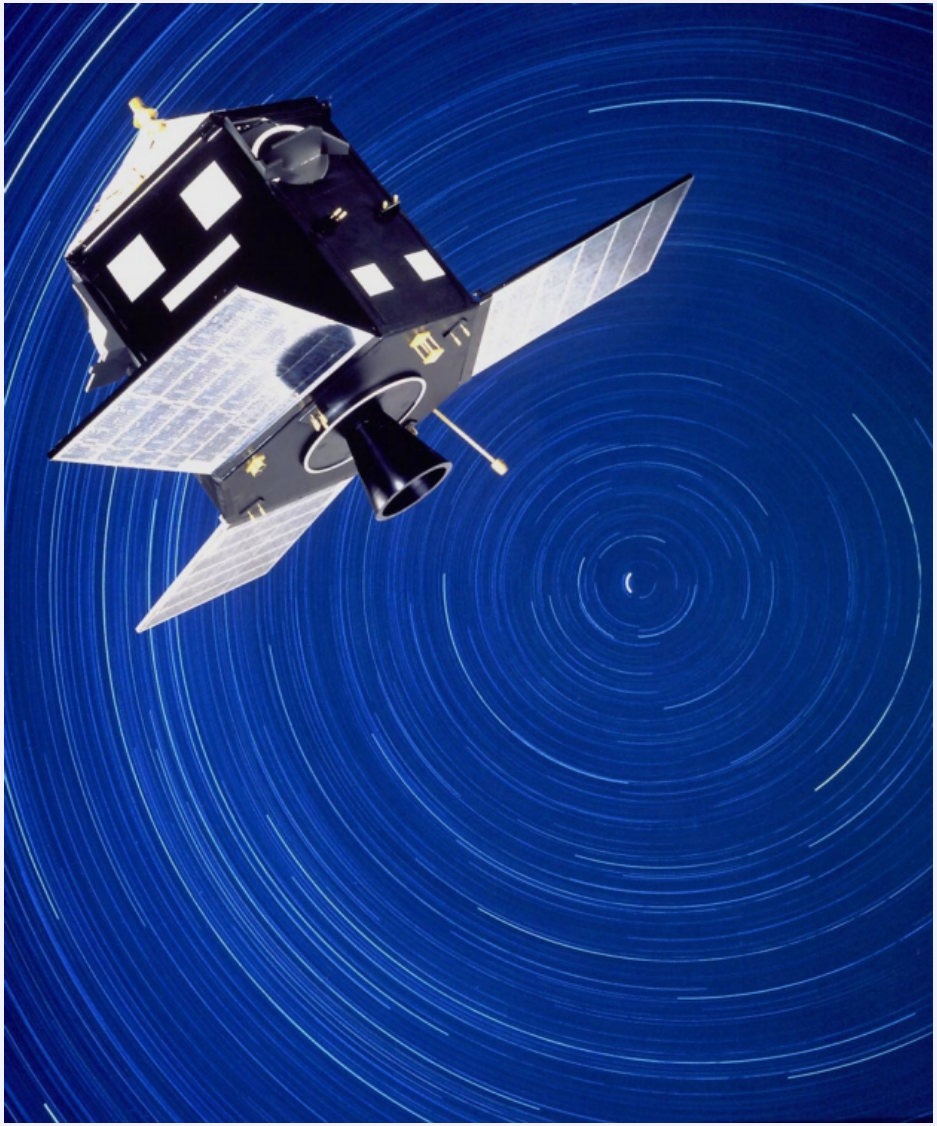


Image credit: ESA

The Hipparcos colour-magnitude diagram

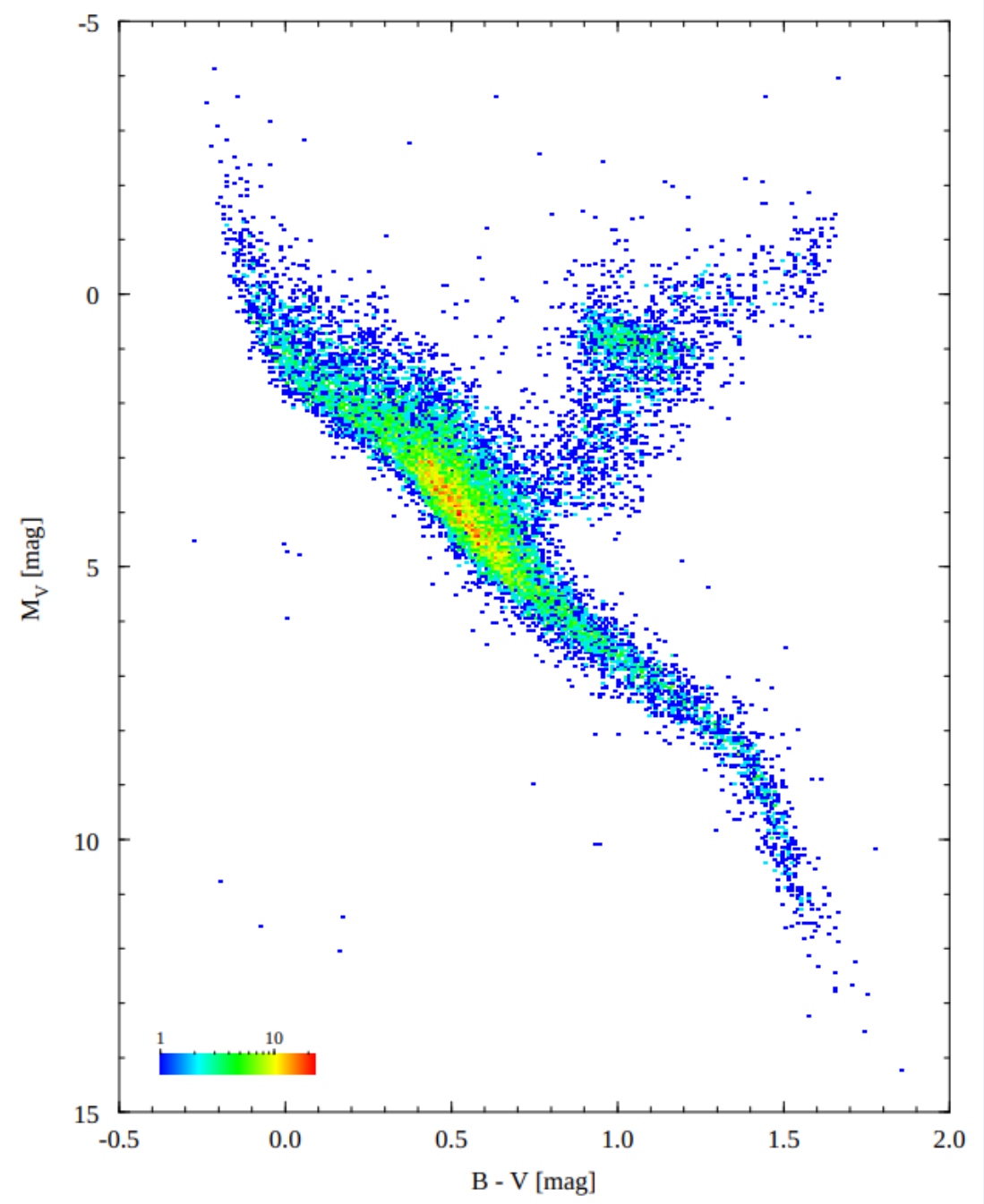
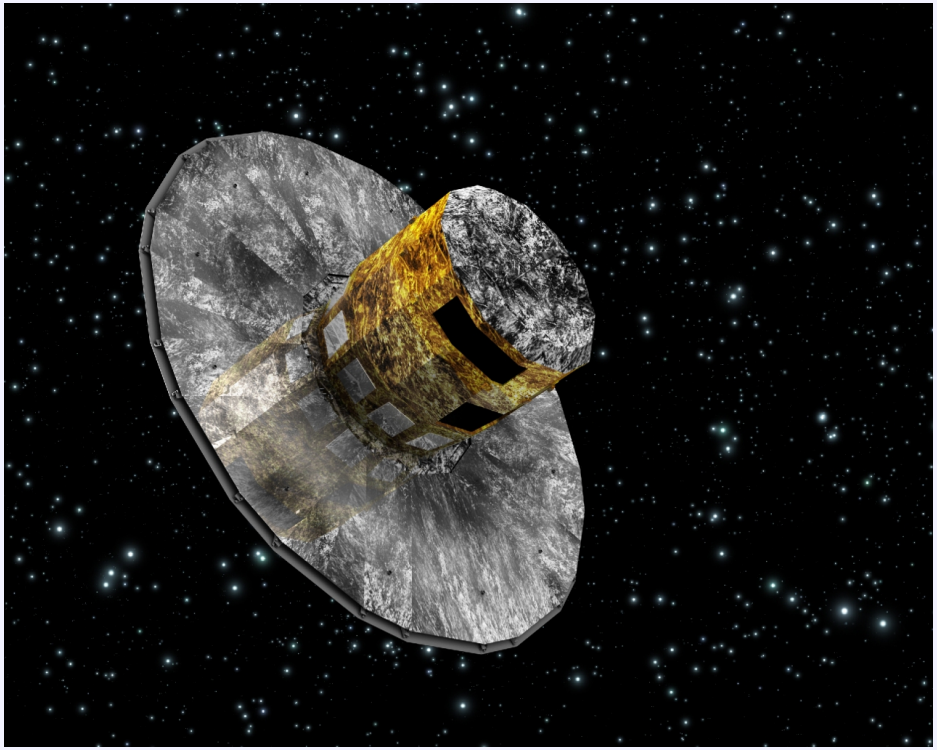


Image credit: ESA

Gaia

- 2013 - 2025
- Astrometric accuracy at the level of 10μas
- Some 100 million stars with distances better than 10%
- More data releases expected



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The Gaia colour-magnitude diagram

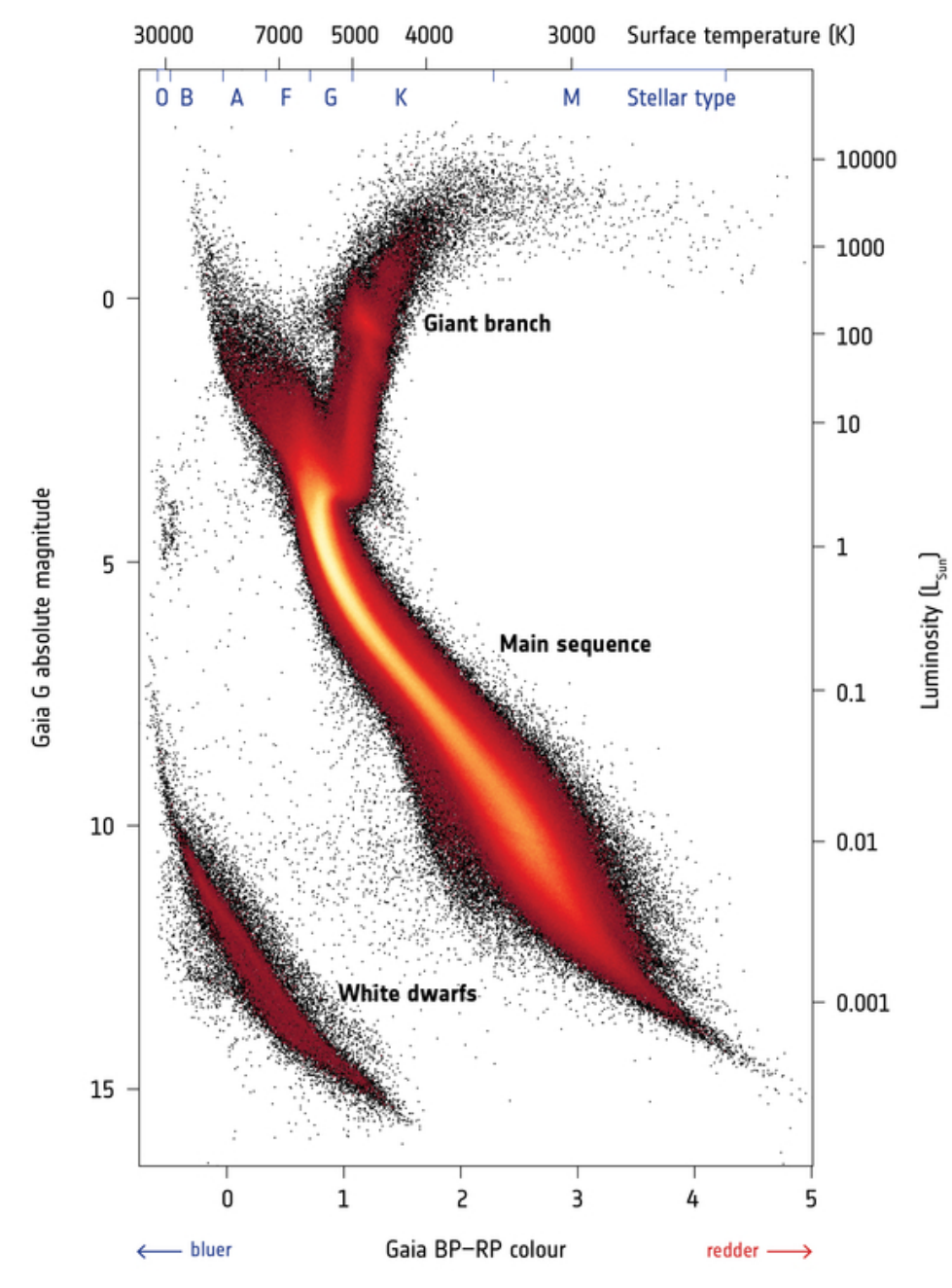

Image credit: ESA

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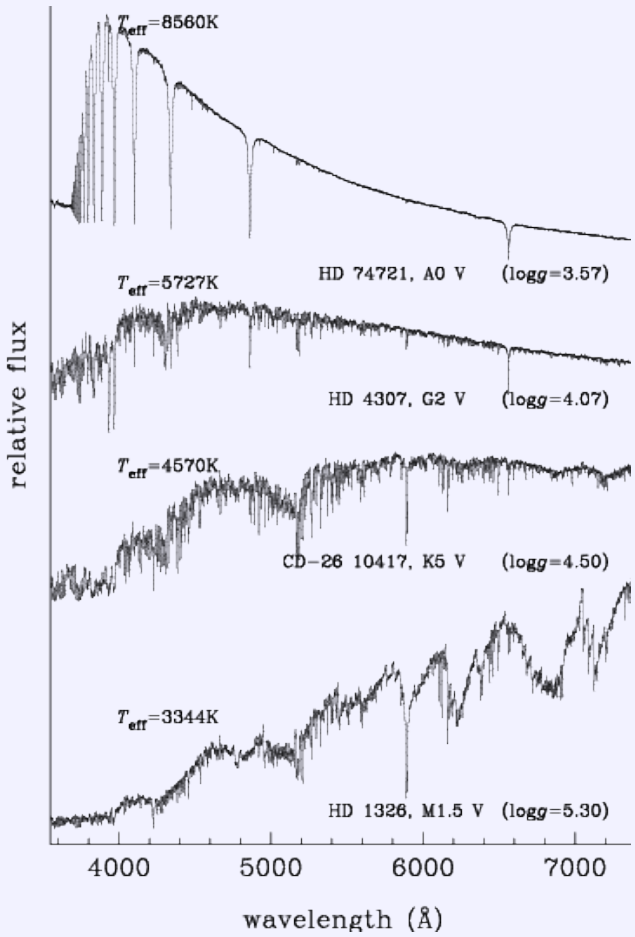
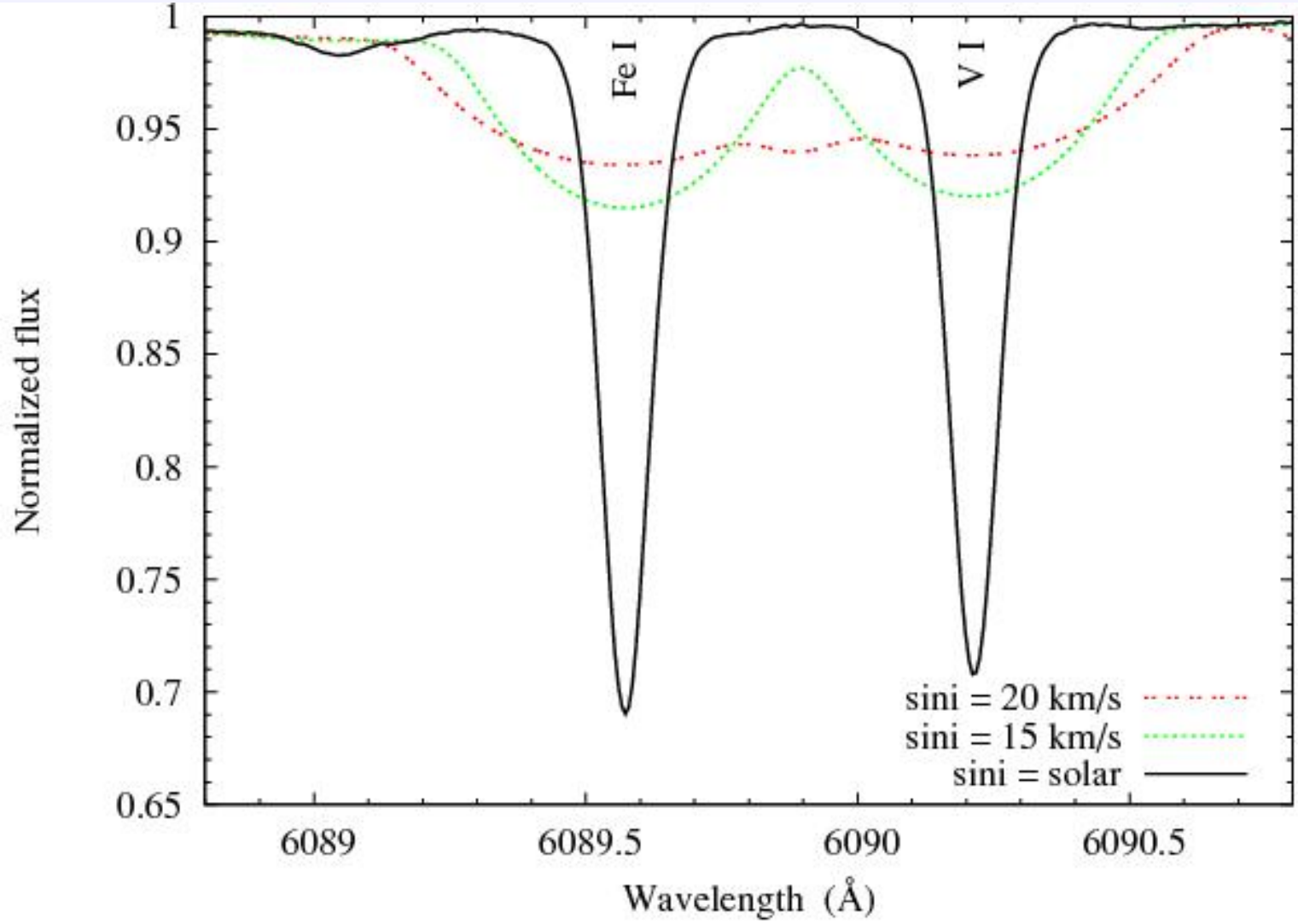
Spectral properties

- Effective temperature T_{eff}
- Surface gravity $\log g$
- Elemental abundances Fe, Mg, Li, ...
- Micro- and macroturbulent velocities
- Minimum rotation rate vsin i

Note this needs knowledge of atomic physics and model stellar atmospheres



Rotation rate from spectra



(Tsantaki et al. 2014)

Stellar abundances - metallicity

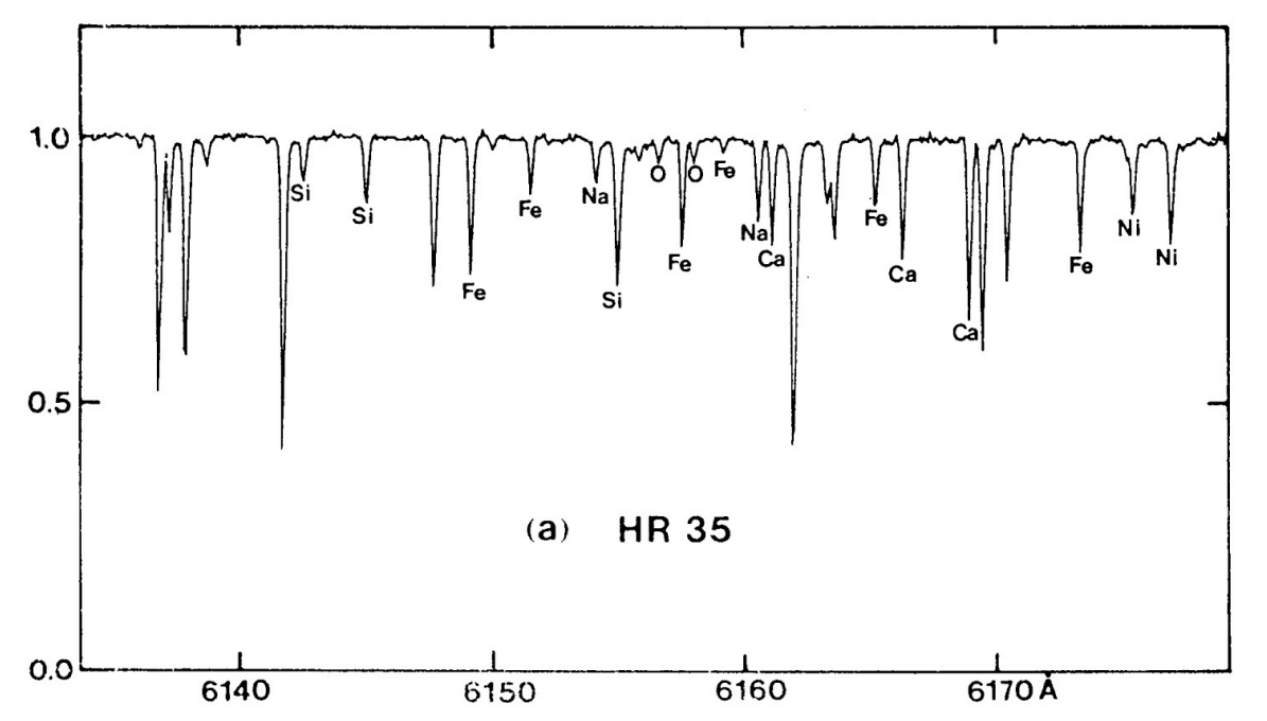
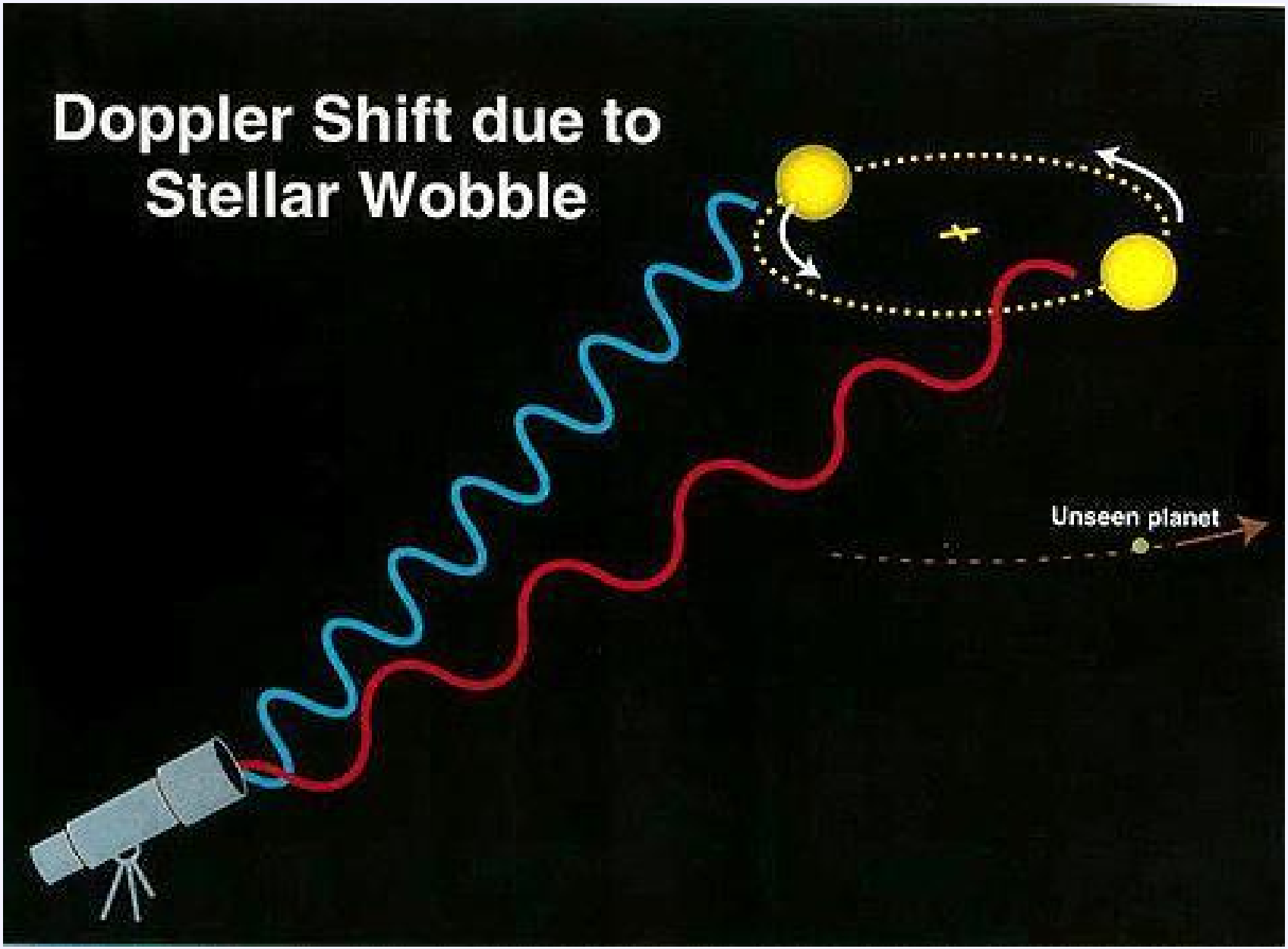


Image credit: JCD

- Most stars consist of H and He
- All other elements classed as "metals"
- Mass fraction of H, He, and metals described by X, Y and Z where

$$X + Y + Z = 1$$

Aside: Spectroscopy as an exoplanet detection technique



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Direct observations of masses and radii

- Masses:
  - Only binary stars allow for direct measurement
  - Through Kepler's third law:

$$M_1 + M_2 = \frac{a^3}{P^2}$$

    with masses in Solar mass, semi-major axis a in AU and orbital period P in years.
- Radii
  - Through eclipsing binaries and photometry
  - Interferometry - only possible for nearby stars
  - Through Stefan-Boltzmann law (since luminosity and effective temperature are measurable with observations)

Masses, radii and ages from stellar models

- Stellar structure and evolution knowledge crucial
- Compare previously mentioned observable properties with stellar models to work out the fundamental parameters

Asteroseismic constraints

- Stars pulsate with different short periods
- These oscillation periods are directly related to internal structure of star
- It can be observed in both brightness/photometry and in RV/spectroscopy
- The different periods per star are directly related to mass and radius and thus age
- Adding asteroseismic observables to stellar models provide most precise parameters
- Side note: asteroseismology and exoplanets are often done together as both need dense and precise photometric surveys

Summary

- Astrometry determines position and distance
- Photometry determines brightness, luminosity, magnitude, colour
- Observational HR diagram through colour-magnitude diagram
- Spectroscopy determines effective temperature, gravity, elemental abundances, rotation rate
- Masses, radii and ages usually determined via stellar modeling
- Asteroseismology for most precise masses and ages

NEXT LECTURE: Equation of state - ideal gas, degenerate gas, radiation pressure